

Tools — Plot

Scientific/terminal plotting rendered off-screen to a text canvas via `ratatui-plt`. Every tool returns the canvas as plain text and mirrors it into output metadata (`plot` plain canvas, `plot_ansi` ANSI-coloured canvas); the TUI renders the coloured canvas inline in the message window. Canvas is clamped to 220x80 cells. All tools are read-only and always visible.

Tool	Description
<code>plot_line</code>	XY line chart from one or more named series of <code>[x, y]</code> points.
<code>plot_scatter</code>	XY scatter with dot markers.
<code>plot_bar</code>	Bar chart from categories + datasets; stacked and horizontal modes.
<code>plot_histogram</code>	Histogram over a sample array; count/density/probability normalisation.
<code>plot_pie</code>	Pie (or donut) chart from labelled slices; auto 8-colour palette cycling.
<code>plot_heatmap</code>	2D grid heatmap; <code>viridis</code> , <code>plasma</code> , <code>inferno</code> , <code>magma</code> , <code>coolwarm</code> colormaps.

Common optional arguments: `title`, `x_label`, `y_label`, `width` (default 80, max 220), `height` (default 20, max 80). Colours accept named colours (`red`, `cyan`, `lightgreen`, ...) or `#rrggbb` hex strings. String-encoded payloads are coerced automatically; malformed input returns an error output rather than crashing.

`plot_line`

Render an XY line chart.

Arguments

Argument	Type	Required	Description	Typical value
<code>series</code>	object or array	yes	One or more <code>{name?, data: [[x,y],...], color?}</code> series	<code>[{"name": "p50", "data": [[1,12], [2,11]]}</code>
<code>title</code> , <code>x_label</code> , <code>y_label</code>	string	no	Chart labels	"Latency"
<code>width/height</code>	integer	no	Canvas cells (80x20 default; max 220x80)	100 / 24
<code>x_grid</code> , <code>y_grid</code> , <code>show_legend</code>	boolean	no	Gridlines and legend toggles	<code>true</code>

Example:

```
plot_line title="Latency" series=[{"name": "p50", "data": [[1,12], [2,11], [3,9]], "color": "cyan"}, {"na
```

plot_scatter

XY scatter plot with dot markers. Same arguments as plot_line.

```
plot_scatter series=[{"name":"samples","data":[[1.0,2.1],[1.5,2.9],[2.0,4.2]]}]
```

plot_bar

Bar chart over category labels with one or more datasets.

Arguments

Argument	Type	Required	Description	Typical value
categories	array	yes	Category labels	["openalex","wikipedia"]
datasets	object or array	yes	One or more {name?, data: [...], color?} datasets	[{"name":"hits","data":[14,22]}]
horizontal	boolean	no	Horizontal bars	false
stacked	boolean	no	Stack datasets	false
title, x_label, y_label, width, height	—	no	Common options	—

Example:

```
plot_bar categories=["openalex","wikipedia"] datasets=[{"name":"hits","data":[14,22]}]
```

plot_histogram

Histogram over a numeric sample array.

Arguments

Argument	Type	Required	Description	Typical value
data	array	yes	Sample values	[3,5,5,7,9,9,9,11]
bins	integer	no	Bin count	20
range	array	no	[min, max] clamp	[0,10]
norm	enum	no	count, density, or probability	"count"
cumulative	boolean	no	Cumulative bins	false

Argument	Type	Required	Description	Typical value
color, title, x_label, y_label, width, height	—	no	Common options	—

plot_pie

Pie (or donut) chart from labelled slices.

Arguments

Argument	Type	Required	Description	Typical value
slices	array	yes	{label, value, color?} entries	[{"label": "rust", "value": 71}]
donut	boolean	no	Render as donut	false
radius	number	no	Donut hole ratio [0,1)	0.35
percentages, labels	boolean	no	Toggle percentage and label overlays	true
title, width, height	—	no	Common options	—

plot_heatmap

2D grid heatmap.

Arguments

Argument	Type	Required	Description	Typical value
grid	object	yes	{values: [[...]]} with optional x/y axis value arrays	{"values": [[0,1],[2,3]]}
colormap	string	no	viridis, plasma, inferno, magma, coolwarm	"viridis"
colorbar, show_values	boolean	no	Colour scale bar and cell value overlays	true / false
title, x_label, y_label, width, height	—	no	Common options	—

Example:

```
plot_heatmap grid={"values":[[0,1],[2,3]]} colormap="viridis"
```